V.A.M.P. | Complete prototype wiring

Sheet 1: power, motor drivers and servo. Sheet 2: breadboard, multiplexer and sensors. All GPIO = BCM.

POWER - direct wires / distribution connectors; keep load current off the breadboard

3S LiPo
11.1 V nominal
12.6 V charged
Regulator inputs only

Existing BEC -> regulated 5 V
Input must accept a full 3S battery

Separate motor buck -> 6 V*
*Use the actual motor rated voltage

+5V: Pi 5V input, servo V+, MQ-2 VCC
Pi power and servo power each get their own direct branch.

VM: DRV1 + DRV2 VM; optional DRV3 VM
If no motor buck: 5 V motor option only if supply current is adequate.

GND: join regulator outputs, Pi, drivers, servo and sensors. Pi 3V3 supplies the sensor rail on sheet 2.

Raspberry Pi
3.3 V logic / BCM GPIO
+5V input
GND

GPIO5
GPIO6
GPIO12
GPIO13
GPIO20
GPIO21
GPIO22
GPIO23
GPIO17
GPIO2 / 3 / 26 -> sheet 2

DRV1 - DRV8833 breakout
VM
GND
STBY*
AIN1
AIN2
BIN1
BIN2
AO1
AO2
BO1
BO2

DRV2 - DRV8833 breakout
VM
GND
STBY*
AIN1
AIN2
BIN1
BIN2
AO1
AO2
BO1
BO2

Rear LEFT / left1
Vertical FRONT / left2
Rear RIGHT / right1
Vertical REAR / right2

Gimbal servo SIGNAL
V+ = +5V GND = common GND

***STBY: keep enabled. If not already pulled HIGH on the carrier, use 3V3 after checking its pull-up.**

OPTIONAL CRAWLER / MANIPULATOR

DRV3: GPIO16 -> AIN1, GPIO19 -> AIN2, AO1/AO2 -> left3; GPIO24 -> BIN1, GPIO25 -> BIN2, BO1/BO2 -> right3.
DRV3 uses the same VM / GND / STBY arrangement. Submarine mode leaves left3 and right3 off.
Arm signal GPIO27; gripper signal GPIO18. Each servo: +5V and GND direct; only one PWM active at a time.

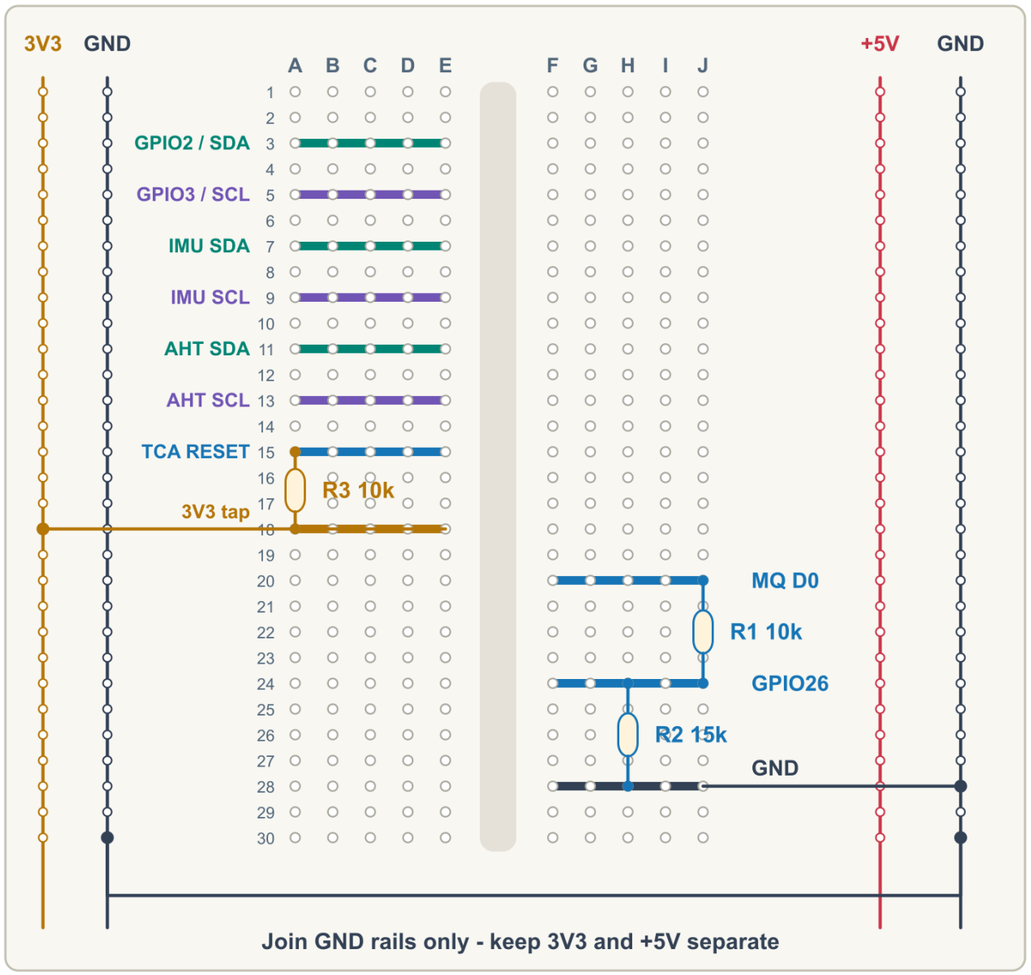
Camera -> CSI ribbon port. PC -> Ethernet. No motor lead goes to GND. NC pins stay unused. Use one Pi power source.

V.A.M.P. | Breadboard signal patch panel

Use labelled jumper wires to each breakout. Hole names such as A3 are breadboard coordinates, never Pi header positions.

Use rows 1-30 of a standard board. A-E are joined across each row; F-J are joined separately.

The centre trench isolates the two halves. Rail strips run vertically; bridge any midpoint split in each strip.



MQ-2 divider circuit

D0 -> 10k -> NODE -> GPIO26

NODE -> 15k -> GND

Ideal 5 V HIGH becomes 3.0 V at NODE.

Carrier pull-ups can reduce HIGH further; measure it.

RAIL FEEDS + Pi JUMPERS

Pi 3V3 -> left 3V3 rail; Pi GND -> left GND rail.

BEC +5V -> right +5V rail; BEC GND -> right GND rail.

Pi GPIO2 -> A3; GPIO3 -> A5; GPIO26 -> F24 (divider node).

Only sensor supply current uses these rails; Pi/servo/motors use sheet 1.

Each separate hole in a rail takes one wire. Check rail continuity first.

TCA9548A - upstream + downstream routing (address 0x70)

VCC -> 3V3 rail; GND + A0 + A1 + A2 -> GND rail.

SDA -> E3; SCL -> E5; RESET / RST -> E15.

SD0 -> A7; SC0 -> A9. SD1 -> A11; SC1 -> A13.

R3: A15 to A18. Jumper E18 to 3V3 rail. RST is held HIGH.

R3 can be omitted if the carrier already has its own 3V3 reset pull-up.

Channels 2-7 stay unused. GPIO2/3 drive the upstream bus only.

HW-123 / MPU-6050 - channel 0 (0x68)

VCC -> 3V3; GND + AD0 -> GND.

SDA -> E7; SCL -> E9. INT / XDA / XCL unused.

Keep the IMU still for its startup gyro calibration.

AHT10 - channel 1 (0x38)

VCC -> 3V3; GND -> GND; SDA -> E11; SCL -> E13.

Place away from the MQ-2 heater; measure ambient air.

MQ-2 breakout - digital output only

VCC -> +5V rail; GND -> GND rail; D0 / DO -> F20.

R1 10k: J20 to J24. R2 15k: H24 to H28.

J28 -> GND rail; F24 -> Pi GPIO26. A0 / AO unused.

Check the node voltage with GPIO26 disconnected: valid HIGH, <= 3.3 V.

READ THIS BEFORE POWERING

A3 / E3 etc. refer ONLY to this breadboard. Breakouts stay beside it; wire by their printed pin labels.

GPIO3 is the clock, NOT the 3V3 supply. Never connect raw MQ-2 D0, +5V or VM to a Pi signal GPIO.

If a downstream I2C branch lacks pull-ups: add 4.7k from its SDA to 3V3 and from its SCL to 3V3.

This is a proposed layout for your default map. Follow config.local.json if you already changed GPIO assignments.

3V3 / +5V / VM are different positive nets. All GND nets share one reference. Never feed raw 3S to DRV8833 VM.